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Common ownership along supply chain and trade credit: Evidence from China

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ABSTRACT

This paper explores the impact of common ownership along supply chain on trade credit based on customer-supplier relationships of Chinese listed firms. The results show that common ownership along supply chain can increase the trade credit. This effect is achieved through two channels: alleviating information asymmetry between customers and suppliers and participating in firm decision making. The effect of common ownership along supply chain on improving trade credit is more significant for non-state owned firms in China. This study provides a reference for equity linkages and financial governance in the supply chain.

1. Introduction

Trade credit plays an important role in making payments when customers purchase goods or services from their suppliers and is often considered as an important source of informal financing for customers (Fisman and Love, 2003; Levine et al., 2018). Earlier literature examined the impact of factors such as financial development (Fisman and Love, 2003), ownership structure (Ge and Qiu, 2007), supplier competition (Chod et al., 2019), and executive-supplier ties (Kong et al., 2020) on trade credit. However, there is less literature examining trade credit from the perspective of equity ties in customer-supplier relationships. This paper extends the line of research in this area by examining the impact of common ownership between customers and suppliers on the aspects of firms' access to trade credit.

Since suppliers provide trade credit without collateral from the customer and without guarantees from third parties or financial intermediaries (Costello, 2013; García-Teruel et al., 2014), then firms are exposed to information asymmetry and moral hazard problems. Research on firm common ownership that has emerged in recent years has found that when firms are linked by common ownership, it can mitigate information asymmetries (Kang et al., 2018) and also improve supervisory efficiency and enhance firms' willingness to disclose (Park et al., 2019). Therefore, I hypothesize that there will be an impact on trade credit when common ownership occurs between suppliers and customers, and verify this empirically.

I study the impact of common ownership along supply chain on trade credit using a sample of listed firms in China, and my main contributions are as follows. First, I provide a new perspective for studying trade credit. Second, existing literature on common ownership mainly studies common ownership of firms in the same industry (Azar et al., 2018; He and Huang, 2017; Lewellen and Lower, 2021; Li and Liu, 2023), and I expand the research related to vertical common ownership from a supply chain perspective. Third, I enrich the literature on supplier-customer linkages; previous literature emphasizes social ties between suppliers and customers

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(Kong et al., 2020), while my study documents the impact of equity ties on supply chain relationships.

2. Data and research design

2.1. Data sources

My sample consists of firms listed on the Shanghai and Shenzhen stock exchanges during the sample period from 2009 to 2020. My data sources are two. First, I obtain firms' financial data and data on firms' top five suppliers and customers from the CSMAR database (a database widely used when studying Chinese firms). Second, I obtain information on the top ten shareholders of a firm from the firm's shareholder database. I identified suppliers and customers with the same shareholders by comparing their shareholder information, and defined them as customer-supplier relationships with common ownership in the supply chain.

2.2. Measurement of common ownership along supply chain

Conceptually, common ownership along supply chain indicates the extent to which suppliers and customers are held by the same shareholder, and I use three methods to measure common ownership along supply chain. First, a dummy variable *cos*1, indicating whether customer i has a common shareholder with its suppliers, takes the value 1 if it does, and 0 if customer i does not have a common shareholder with any of its suppliers.

Second, the product of the equity of the common shareholders of the supplier and the customer firm is calculated (Lewellen and Lower, 2021) and the product of all common shareholders is aggregated as follows:

$$pair - cos2_{i,j} = \sum_{k=1}^{N} common_{i,k} \times common_{j,k} \tag{1}$$

where $common_{i,k}$ and $common_{j,k}$ are equal to the percentage ownership of common shareholder k in customer i and supplier j. The use of the product of ownership stakes rather than the arithmetic mean lies in the fact that the product incorporates the symmetry between $common_{i,k}$ and $common_{j,k}$, the higher the degree of similarity between supplier and customer equity, the higher the value of *cos*2. This customer-supplier level measure is then aggregated at the customer level to form a customer level measure of:

$$cos2_i = \sum_{j=1}^{N} weight_{i,j} \times cos_{i,j} \tag{2}$$

$weight_{i,j}$ denotes the share of customer i's purchasing share from supplier j.

Third, referring to the approach of Kostovetsky and Manconi (2020), the common ownership along supply chain is defined as

$$pair - cos3_{i,j} = \min\{common_{i,k}, common_{j,k}\} \tag{3}$$

That is, the minimum value between the proportion of common shareholders holding *i* customer shares and the proportion holding *j* supplier shares, also contains symmetry. After that, the same is weighted and aggregated to the client firm level according to Eq. (2).

Table 1
Variable definitions.

Variable type	Variables	Variable symbol	Definitions
Dependent variables	Trade credit	*TC*	Ratio of accounts payable to total assets
Independent variables	Common ownership along supply chain	*cos1*	Whether the customer and the supplier have common shareholders
		cos2	The product of the shareholding of common shareholders, summed by the share of customer purchases
		cos3	The minimum value of the common shareholder equity share of customers and suppliers, summed by the share of customer purchases
Control variables	Firm size	*size*	Logarithm of total firm assets
	Firm age	*lnage*	Logarithm of firm establishment years plus one
	Percentage of fixed assets	*ppr*	Total fixed assets over total assets
	Return on Assets	*roa*	Total net earnings over total assets
	Total earnings before interest and taxes	*ebit*	Total earnings before tax and interest over total assets
	Firm cash	*cash*	Total operating cash flows over total assets
	Leverage	*lev*	Total liabilities over total assets
	R&D Expenditure	*rd*	R&D expenditure as a percentage of total assets

Notes: This table presents the definition of key variables and firm-level controls.

2.3. Model design

I construct model (4) to test the effect of common ownership along supply chain on firms' trade credit,

$$TC_{i,t} = \alpha + \beta_1(cos_1 / cos_2 / cos_3) + \sum \beta_j controls_{i,t} + \sum firm_i + \sum year_t + \mu_{it} \quad (4)$$

Measurement of trade credit($TC_{i,t}$). Referring to existing studies, trade credit is defined as total accounts payable divided by total assets. In addition, for robustness testing, I measure trade credit in four other ways. First, the variable *L_TC* is created, which is total accounts payable divided by total liabilities (Fisman and Love, 2003). Second, use (*NET_TC*) equal to net accounts payable-accounts receivable divided by total assets (Liu et al,2016). Third, only the financed portion of accounts payable (*F_TC*) (accounts payable with maturity > 1 year) is used, measured as a percentage of total assets (Ge and Qiu,2007).

To be consistent with the literature, we follow Ge and Qiu (2007), Wu et al. (2014) and Liu et al. (2016) in introducing control variables to capture additional determinants of trade credit. Detailed definitions of all variables used in this study are presented in Table 1. Fourth, to exclude the effect of industry characteristics, the excess of trade credit relative to the industry average is used as the dependent variable.

3. Empirical results and analysis

3.1. Descriptive statistics analysis

The descriptive statistics values of the variables are shown in Table 2. I find, first, that common ownership along supply chain is prevalent in my sample with a mean value of 0.144 for *cos*1, indicating that 14.4% of the customer sample has common ownership with at least one supplier. Second, there is a significant difference in the standard deviation and extreme values of common ownership along supply chain among the sample firms. The minimum value of *cos*3 is 0, while the maximum value is 0.143, which indicates that common ownership along supply chain is sufficient to have an impact on firms' behavior. Third, the firms' trade credit differs significantly with a minimum value of 0 at a mean of 0.0893, while the highest customer trade credit reaches 0.493.

3.2. Baseline regression analysis

Table 3 shows the baseline results of the regression using model (4). In column (1), the coefficient of the dummy variable *cos*1 is 0.0172 and is statistically significant at the 1% level, indicating that customers who have common shareholders with at least one supplier will receive an additional 1.72% of total assets in trade credit. The coefficients of *cos*2 and *cos*3 in columns (2) and (3) are both significantly positive, indicating that common ownership along supply chain increases the level of trade credit of the firm and enables customers to obtain more trade credit from their suppliers.

In addition, the coefficients of the control variables are in line with my expectation. The coefficient of *size* is negative and significant, indicating that larger firms receive less trade credit (Ge and Qiu, 2007). The estimated coefficients of cash held by firms (*cash*) and financial leverage (*lev*) are both positive, which is consistent with the results of Liu et al. (2016).

3.3. Robust tests

Table 4 shows the regression results of the robustness tests, using the other three different measures of trade credit. When using other measurements for the dependent variable, all coefficients of *cos*1 and *cos*3 in Table 4 are statistically significant positive and the coefficient of *cos*2 on *NET_TC* is positive and significant, which confirms the robustness of my findings.

Table 2
Summary statistics.

Variable	N	Mean	SD	Min	p25	p50	p75	Max
TC	2739	0.0893	0.0735	0	0.0360	0.0722	0.129	0.493
cos1	2739	0.144	0.351	0	0	0	0	1
cos2	2739	0.000345	0.00356	0	0	0	0	0.122
cos3	2739	0.000766	0.00583	0	0	0	0	0.143
size	2739	22.95	2.266	16.76	21.40	22.46	24.05	31.14
lnage	2739	2.116	0.879	0	1.609	2.303	2.833	3.401
ppr	2739	0.226	0.181	4.60e-05	0.0819	0.184	0.334	0.937
roa	2739	0.0372	0.0923	-2.746	0.0129	0.0363	0.0670	0.464
ebit	2739	0.0526	0.0943	-2.682	0.0296	0.0521	0.0808	0.568
cash	2739	0.189	0.147	0	0.0855	0.149	0.249	0.993
lev	2739	0.386	0.260	0	0.156	0.384	0.597	0.970
rd	2739	0.00765	0.0777	0	0	0	0	4.554

Table 3
Baseline regression results for common ownership along supply chain and trade credit.

Variables	(1) TC	(2) TC	(3) TC
cos1	0.0172***		
	(0.00248)		
cos2		0.568*	
		(0.328)	
cos3			0.579***
			(0.167)
size	-0.00884**	-0.00958**	-0.00984**
	(0.00388)	(0.00386)	(0.00385)
lnage	-0.00775**	-0.00633*	-0.00674*
	(0.00353)	(0.00353)	(0.00351)
ppr	-0.00603	-0.00338	-0.00256
	(0.0223)	(0.0218)	(0.0221)
roa	-0.0837	-0.0934	-0.0928
	(0.0914)	(0.0899)	(0.0897)
ebit	0.0459	0.0569	0.0562
	(0.0867)	(0.0854)	(0.0851)
cash	0.101***	0.0978***	0.0984***
	(0.0120)	(0.0120)	(0.0120)
lev	0.0721***	0.0719***	0.0717***
	(0.00845)	(0.00844)	(0.00845)
rd	-0.0718	-0.0735	-0.0669
	(0.0663)	(0.0633)	(0.0590)
Constant	-0.0419	-0.0604	-0.0660
	(0.0836)	(0.0833)	(0.0830)
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	2739	2739	2739
R-squared	0.185	0.150	0.155

Notes: The dependent variable is trade credit, and cos1, cos2, and cos3 denote common ownership along supply chain variables. Each regression includes a firm fixed effect for the year fixed effect. Also, we clustered the standard errors by firm. The clustering robust standard errors are shown in parentheses. ***, **, and *, denote statistical significance at the 1%, 5%, and 10% levels, respectively (same below).

3.4. Endogeneity

Considering that the existence of common ownership ties between customers and suppliers may not be exogenous, my results may be influenced by potential endogeneity, especially reverse causality. For example, supply chain relationships where the same shareholder has a greater probability of holding a higher level of trade credit because the supplier-customer relationship is more stable. In this subsection, I use two approaches to address the issue of potential endogeneity. First, the dependent variable is used for period t+1. Second, the 2SLS model with instrumental variables is used to test the model.

Table 5 shows the regression results using the dependent variable in period t+1. After replacing the dependent variable with the trade credit *fTC*in period t+1, the coefficients of the three variables of common ownership along supply chain remain positive and significant, indicating that common ownership along supply chain increases trade credit in period t+1, further demonstrating the robustness of my results.

For the selection of instrumental variables, I am inspired by Schmidt and Fahlenbrach (2017), Gao et al. (2019) and Chen et al. (2023) to use the entry or exit of a firm's suppliers into or out of the constituent stocks of the CSI 300 index, which contains 300 stocks that are " blue-chip" firms, which are more likely to attract investor interest. Belonging to the CSI 300 index may change the indicator that suppliers and customers have a common ownership link, but it does not have a direct impact on a firm's trade credit. Based on the performance and stock price performance of the firms each year, the CSI 300 index adjusts its constituents. Based on this background, I define two dummy variables and, which indicate whether a firm's suppliers entered and exited the constituents of the CSI 300 index in the previous period, respectively. And use these two dummy variables to predict common owernership along supply chain.

The results are shown in Table 6. In the first step of the regression, the coefficient of *in*300 is significantly positive, while *out*300 is not significant, indicating that the entry of suppliers into the CSI 300 index increases the level of common ownership along supply chain. In the second step of the regression, the coefficients of both *cos*2 and *cos*3 are significantly positive, indicating that my results are not altered by the endogeneity problem.

3.5. The mechanism analysis

In trade in supply chains, contractual incompleteness, information asymmetry, and low levels of trust between firms lead to a tendency for suppliers to be cautious in extending limited trade credit to customers. Common shareholders can effectively mitigate information asymmetries, thus facilitating firms to maintain more stable supply chain relationships (Freeman, 2018). As a hub for

Table 4
Common ownership along supply chain and trade credit: roubusting check.

Variables	(1) L_TC	(2) L_TC	(3) L_TC	(4) NET_TC	(5) NET_TC	(6) NET_TC	(7) F_TC	(8) F_TC	(9) F_TC	(10) lnbTC	(11) lnbTC	(12) lnbTC
cos1	0.0357***			0.0206***			0.00372**			0.0316***		
	(0.00425)			(0.00331)			(0.00189)			(0.00705)		
cos2		0.907			0.962***			0.0697			0.7965***	
		(0.776)			(0.306)			(0.0516)			(0.238)	
cos3			1.017***			0.904***			0.0785*			0.751***
			(0.381)			(0.183)			(0.0454)			(0.0843)
Constant	0.214	0.178	0.167	-0.147	-0.171	-0.180	-0.0894*	-0.0931**	-0.0939**	-7.159*	-7.912*	-7.902*
	(0.145)	(0.146)	(0.146)	(0.139)	(0.141)	(0.140)	(0.0467)	(0.0467)	(0.0468)	(3.934)	(4.039)	(4.043)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	2739	2739	2739	2733	2733	2733	2739	2739	2739	2739	2739	2739
R-squared	0.306	0.273	0.277	0.070	0.049	0.053	0.036	0.031	0.031	0.408	0.386	0.384

Note: Due to space limitations, the coefficients of the control variables are omitted from the regression results in this table (same below).

Table 5
Common ownership along supply chain and trade credit: t+1 period dependent variable.

Variables	(1) fTC	(2) fTC	(3) fTC
cos1	0.0168*** (0.00275)		
cos2		0.518** (0.214)	
cos3			0.499** (0.218)
Constant	0.0605 (0.0822)	0.0430 (0.0822)	0.0383 (0.0820)
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	2713	2713	2713
R-squared	0.074	0.044	0.047

Table 6
Common ownership along supply chain and trade credit: 2SLS.

Variables	(1) cos1	(2) TC	(3) cos2	(4) TC	(5) cos3	(6) TC
in300	0.557*** (0.0423)		0.00171*** (0.000386)		0.00505*** (0.000775)	
out300	0.0776 (0.0583)		1.59e-05 (0.000123)		-0.000104 (0.000351)	
cos1		0.0104** (0.00406)				
cos2				4.148** (1.645)		
cos3						1.414*** (0.516)
Constant	-1.270* (0.710)	-0.0113 (0.0390)	0.00473* (0.00259)	-0.0815* (0.0481)	0.0121 (0.00795)	-0.0789* (0.0444)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	2739	2739	2739	2739	2739	2739
R-squared	0.190		0.029		0.092	
Sargan statistic	2.657		1.737		1.848	
P-val	0.1031		0.1875		0.1740	
Cragg-Donald Wald F statistic	180.549		16.094		72.069	

information and resource flow between firms, common shareholders also accumulate rich management experience and can provide information support for firms' business decisions (Brooks et al., 2018). Common shareholders also enhance firms' willingness to disclose (Park et al., 2019), which can alleviate information asymmetry between suppliers and firms. At the same time, common

Table 7
Common ownership along supply chain and trade credit: The test of Intermediary mechanism.

Variables	(1) TC	(2) TC	(3) TC	(4) TC
cos2_dist	0.124** (0.0593)			
cos3_dist		0.000716 (0.0484)		
cos2_dir			0.187 (0.376)	
cos3_dir				0.561** (0.237)
Constant	-0.0586 (0.0824)	-0.0640 (0.0823)	-0.0404 (0.0820)	-0.0435 (0.0817)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	2739	2739	2739	2739
R-squared	0.153	0.156	0.236	0.239

shareholders can form a positive influence on corporate governance during their participation in the business process and supervise the business decisions of the firm (Kang et al., 2018).

Based on this, I examines two potential channels for common ownership along supply chain to improve trade credit, alleviating information asymmetry and participating in firm decision making.

First, drawing on the existing literature (Li et al., 2015; Ding et al., 2023), we use the geographical distance from the supplier firm to capture the degree of information asymmetry between customers and suppliers. Adding the cross product term $cos2 \times dist$,of common ownership and geographic distance to Eq. (4), if the coefficient of this term is positive, indicates that the impact of common ownership along supply chain on trade credit is greater when information asymmetry is higher, and there exists a channel to mitigate information asymmetry.

Second, appointing directors may be an effective way for common shareholders to participate in firm governance, which in turn helps them to participate directly in making firm investment decisions. Consider that common shareholders have an incentive to induce suppliers to provide larger trade credits to their customers. To test this channel, I define shareholder-appointed directors as "directors who also work for a common shareholder" and manually collect data on directors appointed by common shareholders to suppliers during the sample period. A dummy variable *dir* is constructed to indicate whether the common shareholder appoints a director to the firm's supplier, and the cross product term $cos2 \times dir$ is tested. If the coefficient is positive, it indicates that after the common shareholder appoints a director, the customer receives more trade credit from the supplier, proving that the channel of participation in the firm's decision making holds.

Table 7 shows the regression results of the channel test. The coefficient of the cross product term $cos2 \times dist$ is positive and significant, indicating that the channel of information asymmetry holds. The coefficient of the cross product $cos3 \times dir$ is also positive and significant, indicating that the channel of participation in firm decision making is valid.

3.6. Heterogeneity of firm property rights

In the context of low level of financial development, non-state firms in China have less access to finance compared to state-owned firms. As a result, non-state firms in China receive more trade credit (Ge and Qiu, 2007). Whether supply chain common ownership linkages help non-state firms to obtain more trade credit when suppliers and customers have common shareholders. I divide the sample into state-owned and non-state-owned firms for regressions to test for heterogeneity. The results, shown in Table 8, are positive and significant for the non-SOE group, while the results for SOEs are not significant, indicating that common ownership along supply chain mainly helps non-SOEs to obtain more trade credit.

4. Discussions and conclusion

I studied the impact of customer and supplier common ownership on trade credit from the perspective of supply chain equity linkages and obtained the following conclusions. Common ownership along supply chain significantly increases the size of trade credit. After applying a series of robustness tests such as the replacement trade credit measure and dealing with endogeneity issues, my conclusions remain robust. Further channel tests suggest that common ownership along supply chain enhances firms' trade credit by mitigating information asymmetry between customers and suppliers and dispatching directors to participate in the channel of firm decision making. Heterogeneity tests that distinguish between firm ownership indicate that common ownership along supply chain improves trade credit of non-state firms more significantly.

Based on this, I make two suggestions. First, encourage suppliers and customers to overcome the information asymmetry and inter-firm trust problems in supply chains, such as through common ownership. Secondly, common shareholders should actively explore effective mechanisms to enhance the market value of upstream and downstream firms in the supply chain by virtue of their governance advantages, so as to alleviate the information friction and financing constraints between suppliers and customers. Especially Chinese firms have been integrated into the global supply chain system, and for supply chain common shareholders, their main advantages are information resources and management experience, which should promote the improvement of supply chain governance model of Chinese firms.

For suppliers, the risk of customers' failure to repay all debts should be considered in the case of providing trade credit to customers, especially when market demand is unstable and the credit status of downstream firms is poor (Zhang et al., 2014). In addition, the structure of supply chains is divided into vertical and horizontal, and different structures of supply chains may be affected by common ownership differently, but this paper is limited by data and does not support the distinction between horizontal and vertical supply chains, and expects that subsequent studies can answer this question.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 8
Common ownership along supply chain and trade credit: Heterogeneity of ownership.

Variables	(1) State-owned firms TC	(2) Non-state owned firms TC
cos3	0.929	0.442**
	(0.686)	(0.223)
Constant	-0.0598	0.0128
	(0.0963)	(0.117)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
N	1349	1390
R-squared	0.150	0.199

Data availability

The data that has been used is confidential.

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